Nurullah Said Tuğma
150720823

1) $y = \frac{\lambda L}{d}$

$y = \frac{0.5\times10^{-6} \times 5}{0.4\times10^{-3}} = 6.25\times10^{-3}\ m = 6.25\ mm$

n = 1 to 5 :

first dark spot (n=1) → 6.25 mm from the central bright

second dark spot (n=2) → 12.5 mm from " " "

3'rd dark spot (n=3) → 18.75 mm from " " "

4st dark spot (n=4) → 25 mm from " " "

5th dark spot (n=5) → 31.25 mm from " " "

Nurullah Said Tügmen

150720823

4) * Same responsivity

* Same brightness

$\Phi = \Phi$

$Lum = \eta_{LE} P_o (633\ lm/W)$

$P = I.V$ → electrical power

optical power ? → Luminous efficiency

Luminous efficiency at 480nm → $\overset{blue}{\eta} = 20\ lm/W$

" " at 590nm → $\overset{yellow}{\eta} = 35\ lm/W$

$$\frac{P_{o\,blue}}{\eta_{blue}} = \frac{P_{o\,yellow}}{\eta_{yellow}}$$

* $P_o$ must be same for both LED

$$\frac{P_{o\,blue}}{20} = \frac{P_{o\,yellow}}{35}$$

* Blue LED > Yellow LED

↓ (carrying current)

Lower Luminous efficiency

---

2) Indirect Bandgap: Silicon has indirect bandgap. this means minimum energy required for an electron to transition from the valence band to the conduction band is relatively high, so this is inefficient light emission in the green wavelength range.

- solution → to utilize quantum confinement effects in nanostructures.

Material quality: Si suffers from low material quality in terms of impurity levels and defect densities compared to compound semiconductors.

These impurities and defects can degrade the LED's performance and efficiency

- Solution → improve the material quality of Si by using advanced technologies and purification process can help mitigate this issue,

Nurullah Said Tuğman 150720823

## 3) Advantages

1) Bandgap Engineering: Enables the design of LEDs with specific emission wavelenghts

2) Carrier Confinement: Heterojunction structures confine carriers to specific regions improving carrier recombination and increasing light emission

## Disadvantages

1) Epitaxial Growth Challenges

2) Material Compatibility: Heterojunction structures with different materials can lead to thermal stress, causing material degradation, reduce efficiency and reliability issues.